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constexpr containers and adaptors

This paper proposes marking all containers and adaptors `constexpr` (with exception of `std::hive`). This will make compile time programming much easier by making normal C++ and `constexpr` C++ more similar.

This paper doesn't propose changes in `std::hash`.

Revision history

- [R1](#) → [R2](#): [Removal of `constexpr` for `node\_handle::mapped\(\)` member function](#), Removal of `ceilf` mention in the implementation section as the function is `constexpr` since C++23. [Added discussion about hashing and why it's not included](#), [Added informations about `libstdc++` implementability](#), [Extended discussion about change impact on existing code](#).
- [R0](#) → [R1](#): Added implementation specific problems with `std::less<T*>`.

Introduction and motivation

All necessary `constexpr` functionality required by the containers to be changed is supported by the language. There is no reason for them not to be `constexpr`, and the standard library should support these in order for them to properly serve as vocabulary types.

The following table shows what this paper proposes, and the status quo:

container / adapter	before	after	
<code>std::array</code>	✓	✓	
<code>std::deque</code>	✗	✓	
<code>std::forward_list</code>	✗	✓	
<code>std::list</code>	✗	✓	